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TITLE: ViT-AUDIT: Spectrogram-Based Deepfake Audio Detection Using Vision Transformers

Abstract

With the rise of artificial intelligence, generating fake audio that sounds like a real person has become easier and more convincing. These deepfake audios can be used to trick voice recognition systems, posing serious risks in areas like banking, customer service, and personal security. To address this, our project presents a novel approach to detecting fake audio by transforming audio signals into spectrograms—visual representations that capture pitch and tone variations over time. These spectrotemporal patterns are then analysed using Vision Transformers (ViTs), a deep learning architecture originally designed for image recognition, adapted here for audio forensics. By leveraging the powerful pattern recognition capabilities of ViTs, our system can effectively distinguish between genuine and AI-generated voices. Unlike traditional audio-based models, our vision-based method offers enhanced accuracy by capturing intricate time-frequency dependencies. To improve the system's generalization across varied environments, we also incorporate diverse preprocessing steps, such as background noise filtering and normalization, to prepare the data more effectively for analysis. This ensures our model remains robust even under real-world audio distortions. Our results will demonstrate the efficacy of this approach, providing a robust and scalable solution for safeguarding voice-based systems against synthetic audio threats.

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